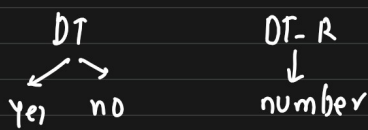


12-07-2026

Agenda: ML-18

- Decision Tree Regressor •
- Ensemble method •
 - Bagging & Boosting
- Bootstrap data •
- Random forest •

DT Regressor predicts a continuous value eg. house price, temp, stock value...



Aspect	classification Tree	Regression Tree
output	class label	Numeric value
Impurity	Entropy / Gini	variance / MSE
leaf prediction	Majority class	Mean / Median of values
split criterion	Information Gain	Variance Reduce

idea behind DT-reg :

$y \rightarrow 10, 20, 20, 80, 90, 100$
 $y_1 \rightarrow 10, 20, 30 \rightarrow$
 $y_2 \rightarrow 80, 90, 100 \rightarrow$ smaller

Dataset:

10 lakh = 1million

ID	Square feet	price (₹ in lakh)
1	800	50
2	850	52
3	900	54
4	1500	90
5	1600	95
6	1700	100

Step:1 Compute parent variance (price)

$$\mu(\text{price}) = 73.5$$

$$\text{var} = \frac{1}{n} \sum (y_i - \mu)^2$$

$$= \frac{1}{6} \sum (50 - 73.5)^2 + (\dots)$$

$$= \underline{\underline{471.9}}$$

Step 2:

for each feature $\rightarrow$ compute below

$\rightarrow$ pick a column / feature

$\rightarrow$ sort the values in ascending order

$\rightarrow$ split point (midpoint between value)

$\rightarrow$ for loop (split point)

$\rightarrow$ we test & compute variance Reduction (VR)

Example:

X1	X2
10	80
20	90
30	100

$$X2 > 85$$



(15, 25)

(85, 95)



VR



VR



VR



VR

$\rightarrow$ highest reduction becomes node

Step 2. calc: —

- square-foot column
- sorted → 800, 850, 900, 1500, 1600, 1700
- split point (825, 875, 1200, 1550, 1650)

split at 1200 :

Square-foot ≥ 1200			
left		right	
800	50	1500	90
850	52	1600	95
900	54	1700	100

$$\begin{aligned}\mu(\text{left}) &\rightarrow 52 \\ \text{var}(\text{left}) &\rightarrow 8/3 \rightarrow 2.67\end{aligned}$$

$$\begin{aligned}\mu(\text{right}) &\rightarrow 95 \\ \text{var}(\text{right}) &\rightarrow 56/3 \rightarrow 18.67\end{aligned}$$

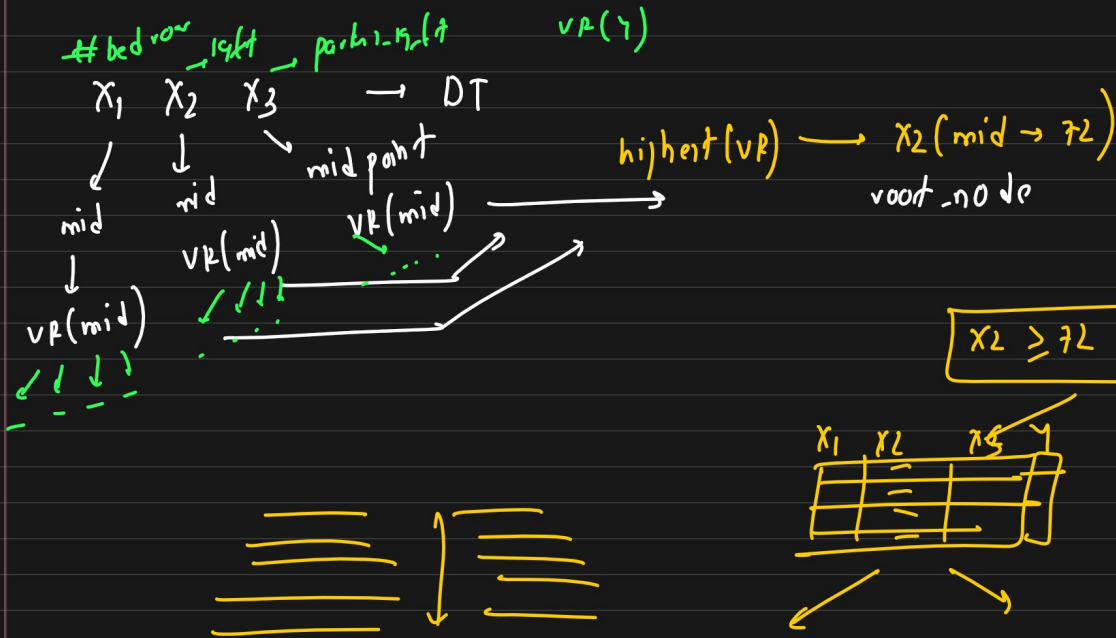
weighted variance:

$$\begin{aligned}\text{var_split}(1200) &= \frac{3}{6} \times 2.67 + \frac{3}{6} \times 18.67 \\ &= 1.335 + 8.335 = 9.67\end{aligned}$$

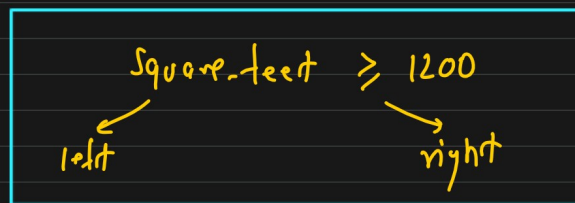
$$\begin{aligned}\text{variance_reduction}(1200) &= 471.9 - 9.67 \\ &= 462.23\end{aligned}$$

let, suppose VR at 1200 is highest

→ that will make $\text{sqft} \geq 1200$ | $\text{sqft} < 1200$ root node



Back to original calculation



800	50	1500	90
850	52	1600	95
900	54	1700	100

Step 1:

for each feature, cal mid point $\rightarrow VP$

midpoint: 825, 875

midpoint: 1550, 1658

split: 825

split: 1550

800, 25	850, 52
900, 54	
$\downarrow$	$\downarrow$
$var(L)$	$var(R)$
$\downarrow$	$\downarrow$
0	1

1500, 90	1600, 95
1700, 100	
$\downarrow$	$\downarrow$
$var(L)$	$var(R)$
$\downarrow$	$\downarrow$
0	6.25

$$w.var = \frac{1}{3} \times 0 + \frac{2}{3} \times 1 = 0.67$$

$$w.var = \frac{1}{3} \times 0 + \frac{2}{3} \times 6.25 = 4.16$$

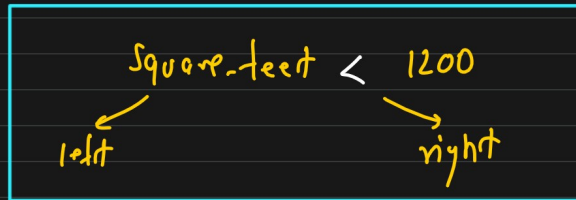
$$VR(825) \rightarrow 2.67 - 0.67$$

$$\rightarrow 2.0$$

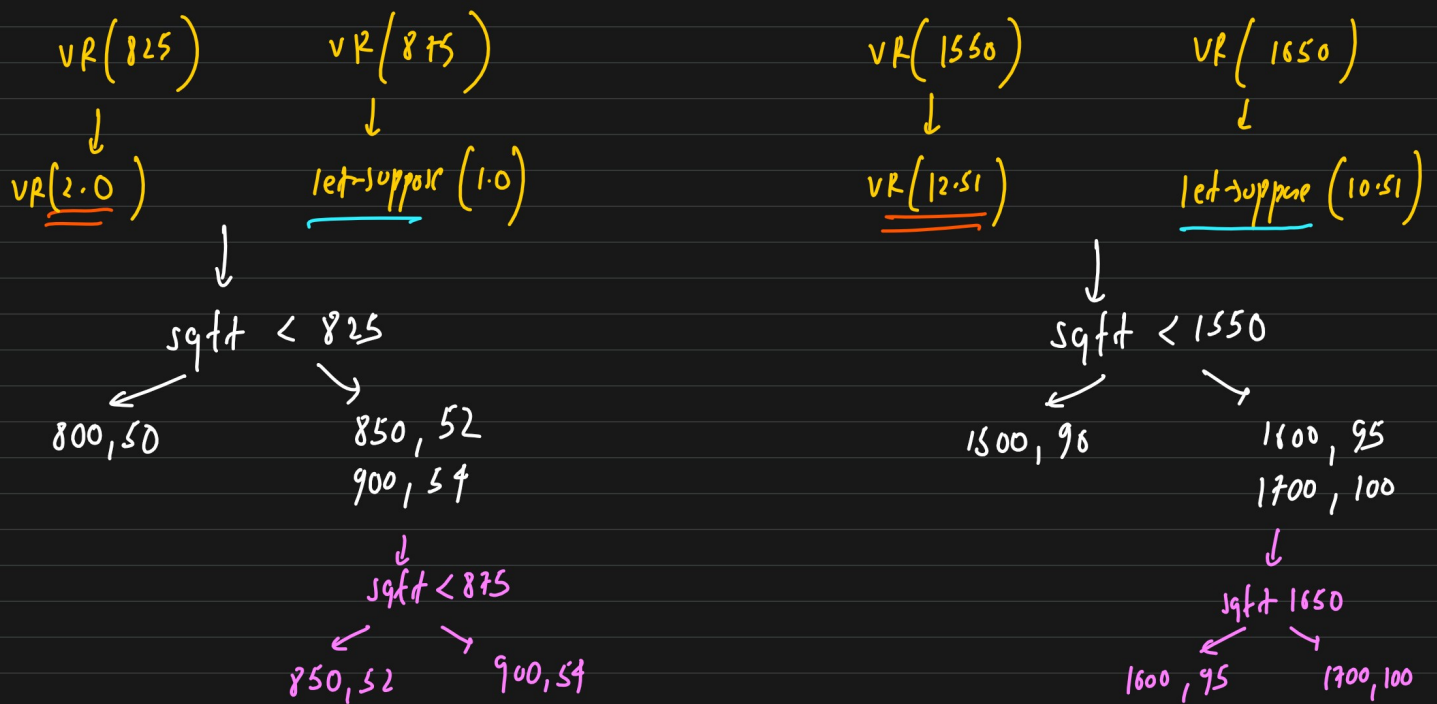
$VR(825) \rightarrow$ let suppose 825 is
the best split

$$VR(1550) = 16.67 - 4.16$$

$$= 12.51$$



800	50	1500	90
850	52	1600	95
900	54	1700	100



Test $\rightarrow$ sqft $\rightarrow$ 1680 $\rightarrow$ price

1650
1686

Overfit:

if all the leaf node are pure, it will lead to overfitting

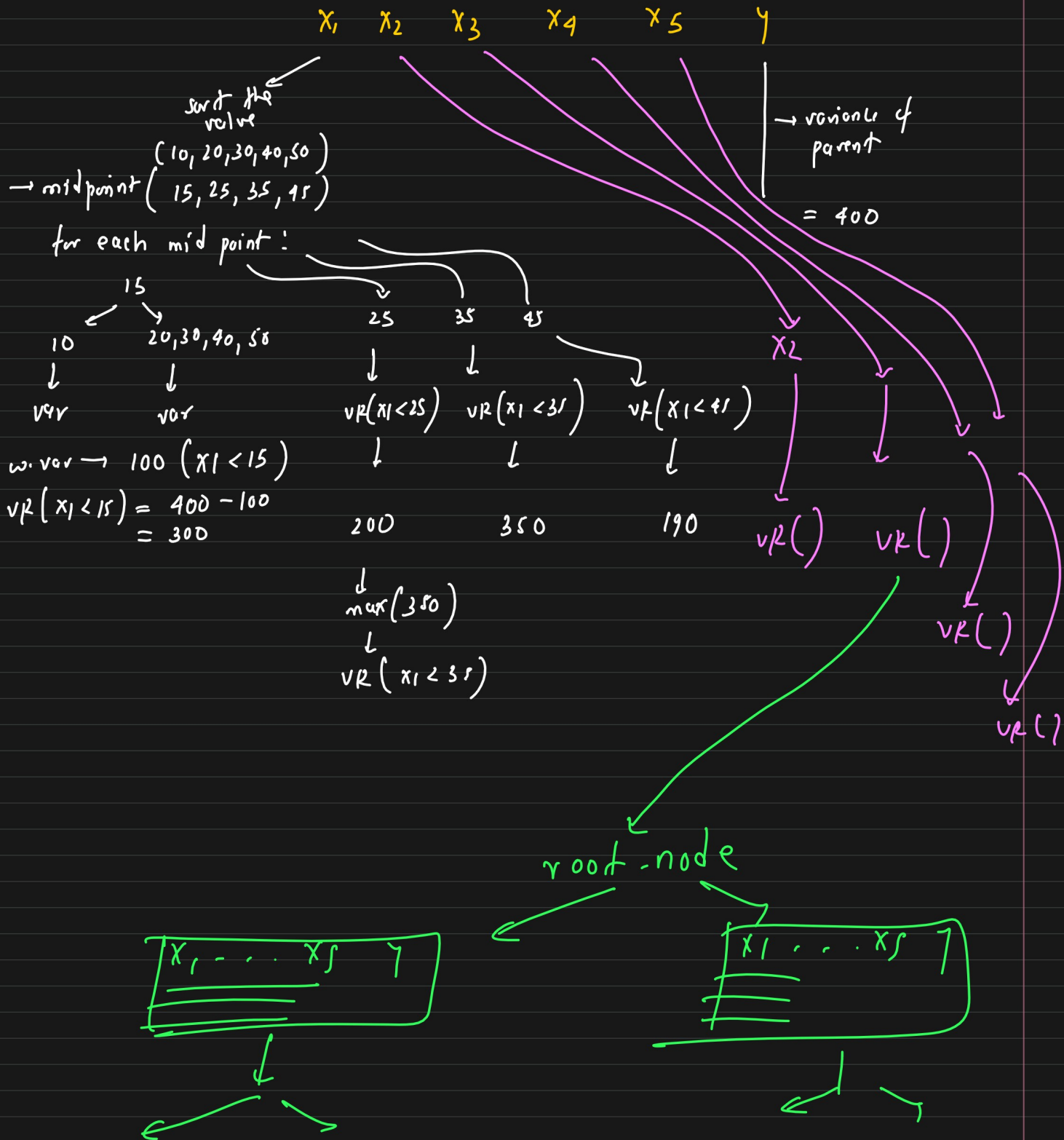
min-sample-split = 2

split (if $\text{len}(\text{data}) > 2 \rightarrow \text{split}$

min-sample-leaf = 1 - Overfit

greater than or equal to 1

→ pre-pruning (early stopping)



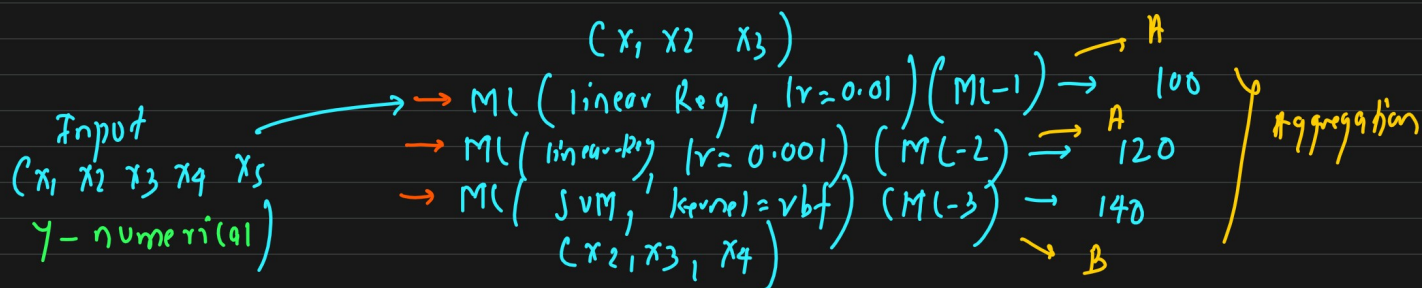
Ensemble method / learning / technique

↓
creating multiple models for a problem statement

↳ Bagging

Random forest (Bagging + Decision Tree)

↳ Bootstrap (not a part of Ensemble)

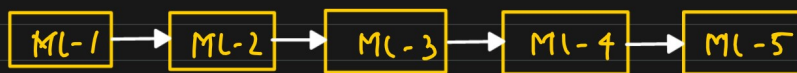


(bagging) → parallel model (independent) classification → Majority
Regression → average, median

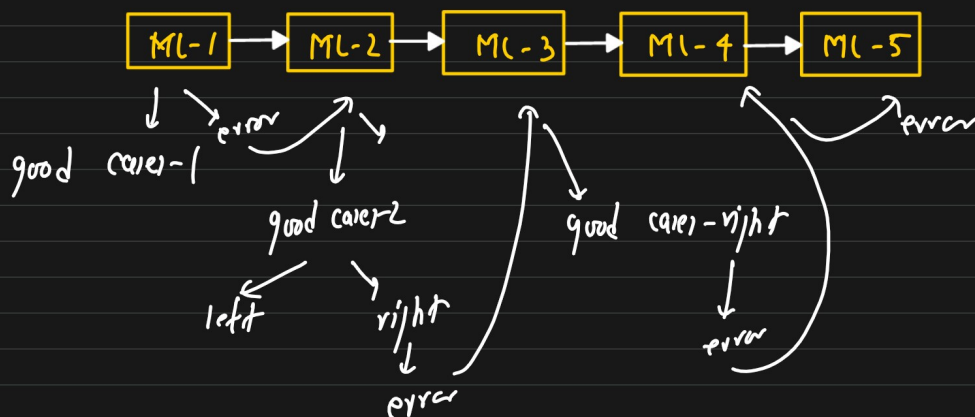
↳ Boosting

↳ xgboost, Gradient Boosting

↳ (boosting + Decision Tree + other)
→ sequential model

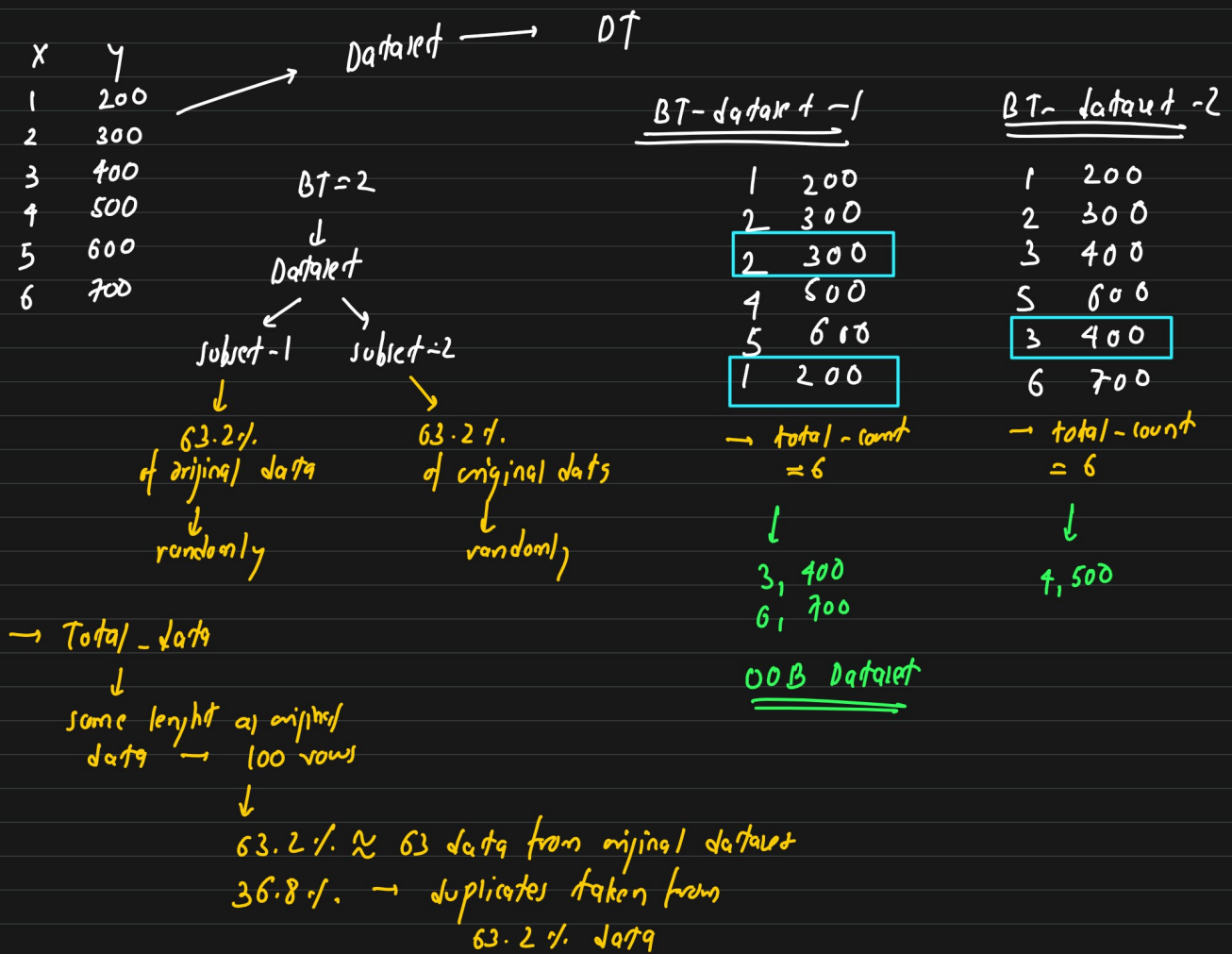


All the model have simple architecture / base-model / simple model

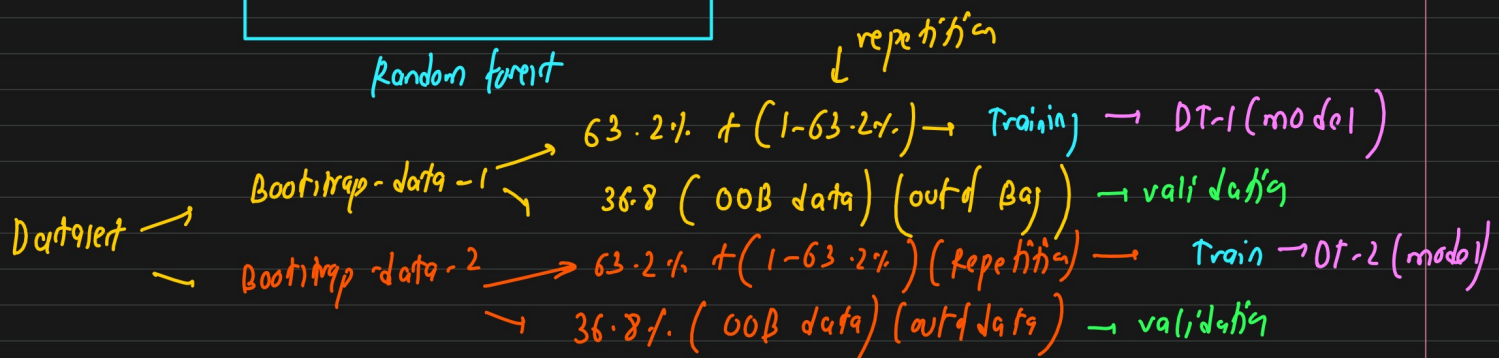
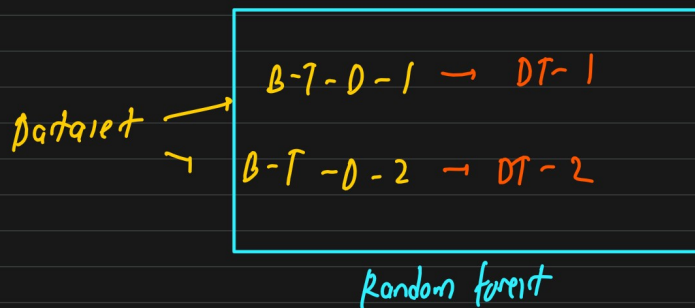


↓
Tree
↓
stumps
(depth = 1, 2, 3)

Bootstrap (Dataset)



Dataset $\rightarrow$ DT $\rightarrow$ 1 complex-DT $\leftarrow$ overfit



Random forest is self-validated model

prob of a specific row never getting picked after N random draws

$$OOB = \left(1 - \frac{1}{N}\right)^N \rightarrow \text{N becomes very large}$$

$$\frac{1}{e} \approx \frac{1}{2.718} \rightarrow 0.3678 \rightarrow 36.8\% \text{ (OOB data)}$$

$$(100 - 36.8\%) \rightarrow 63.2\% \rightarrow \text{bootstrap data}$$

Random forest : (Bagging)

Dataset $\rightarrow$ $n_{\text{estimators}} = 100 \rightarrow$ 100 Bootstrap data $\rightarrow$ DT-1 DT-2 DT-3
 $\rightarrow$ 100 DT BT-Data-1 BT-Data-2 BT-Data-100

$\rightarrow \text{max_feature} = 2$

Bagging + Bootstrap + Decision Tree

x_1, x_2, x_3, x_4

for every node (root/decision)
 $\rightarrow$ allowed to select node from 2 random features

$\rightarrow$ calculate parent variance

$\rightarrow$ root $\rightarrow x_1$

sort values

mid point

x_3

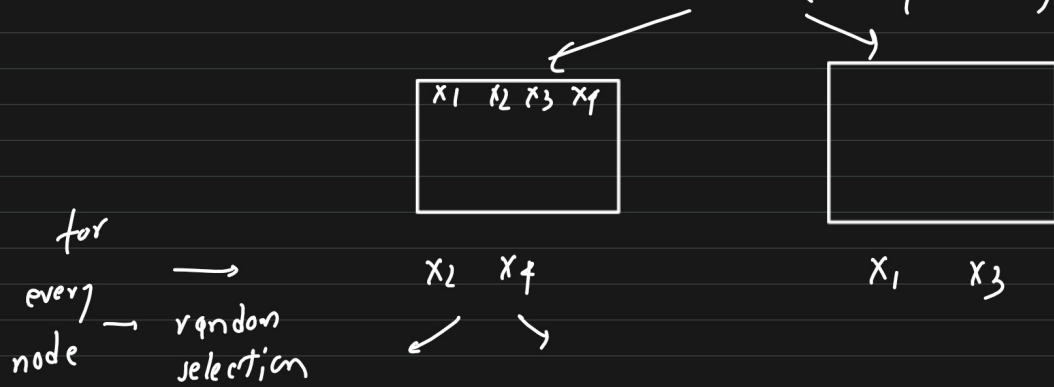
sort values

mid point

for each: variance (left & right) variance (left & right)

$\rightarrow$ find the best mid point using the highest variance reduction

→ root node decision (completed)



Final - prediction:

DT → f_N → model-N
 ↓
 Decision tree index function

DT → f_n → $\hat{y}$

$N = 100$

$$\hat{y}_{RF}(x) = \frac{1}{N} \sum_{i=1}^N f_n(x)$$

model-i

sqft = 1200

bedroom = 3

age = 4

$$f_1(x) = 97.5$$

$$f_2(x) = 93.75$$

$$f_3(x) = 90$$

$$\hat{y}_{RF}(x) = \frac{1}{3} (97.5 + 93.75 + 90)$$

$$= \underline{\underline{93.75}}$$

easy to interpret:—

Decision Tree ✓

Random forest X Black Box



very powerful